

COS: C-Series — Transmission Architecture

Cognitive OS Spec: C-Series

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Abstract

The same content, delivered differently, can produce different outcomes. Transmission failure may not always be a problem of content. The output format itself may influence the result of transmission.

This paper organizes this phenomenon into four output formats: C1 (Structure-first), C2 (Comprehension-first), C3 (Reception-first), and C4 (Minimal Form). These formats do not indicate superiority. Each carries a different distribution of transmission cost.

C-Series is not a model that prescribes the correct way to communicate. It is a model that makes observable the format in which thought content is being output. In doing so, it offers a framework for understanding why the same content can produce different outcomes depending on output format. The C-Series addresses how responses generated by the B-Series are transformed into output formats.

This paper is the third in the COS (Cognitive Operating System) series, and proposes a definition of the transmission strategy module within that framework.

Keywords: C-Series, Transmission Strategy, Output Formats, Communication Cost, Human-AI Interaction, COS

Chapter 1: Why Correct Content Still Fails to Reach

We communicate something to someone every day.

Consider this message: "We will not be moving forward with your proposal."

The same decision might be delivered as "We'll pass on this one," or "We'll keep it in mind for future reference," or "Let us get back to you on that." The underlying content is similar. Yet the way it lands can be entirely different.

Even with the same recipient and the same information, a shift in how something is expressed can produce a significantly different reaction. The accuracy of the content, or the characteristics of the recipient, may not fully account for this variation.

This phenomenon is not limited to human-to-human communication. It appears in business settings, customer support, and dialogue systems alike. When the intended message remains constant but the output format changes, the recipient's response can shift accordingly.

Communication failure may not always be a problem of content. The output format itself may influence the outcome of transmission.

This is distinct from the misalignment in interpretation that B-Series addressed. B-Series examined how we receive what others say. C-Series addresses how we output what we intend to say.

C-Series is a model that classifies the process of converting thought into output format into four types, making it observable.

Chapter 2: What Is C-Series?

2-1. What Is C-Series?

C-Series is the transmission strategy module within COS (Cognitive Operating System).

Its role is not to guarantee the correct way of communicating. It organizes the process of converting thought into output format and breaks down differences in output format into observable structures.

Output format is often perceived as a single act. In practice, however, the same content may be expressed in more than one format.

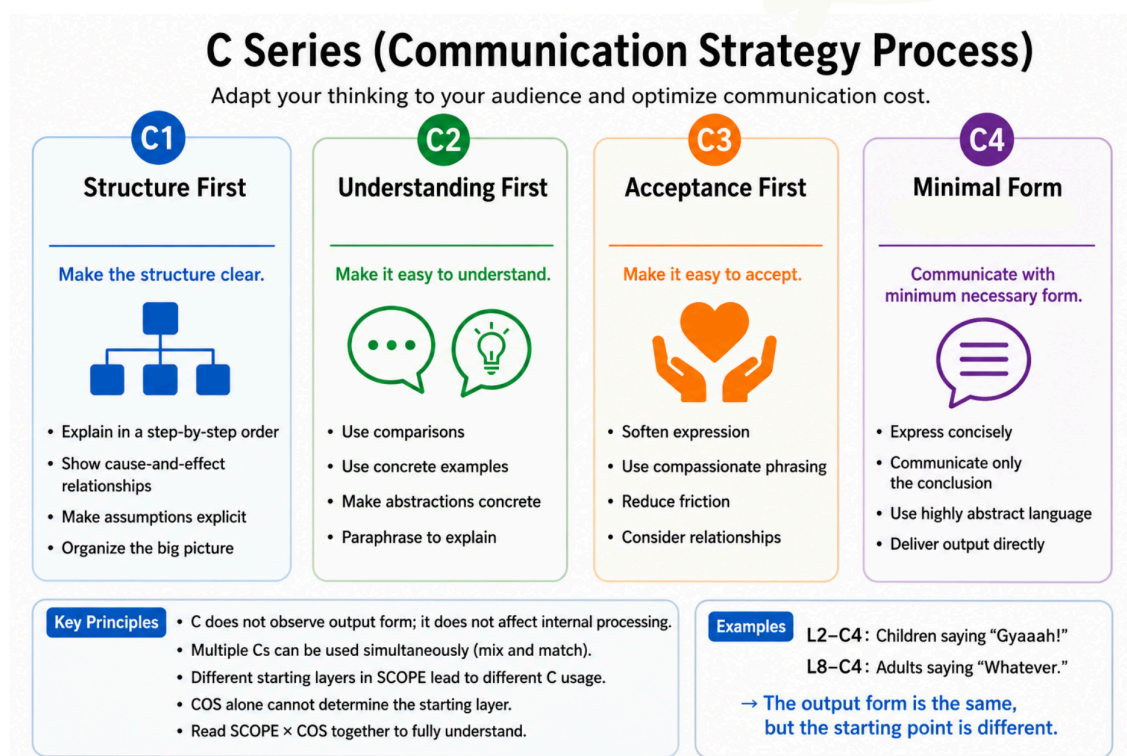
C-Series organizes these formats into four types.

- C1: Structure-first
- C2: Comprehension-first
- C3: Reception-first
- C4: Minimal form

These four formats do not indicate superiority or level of sophistication. Nor does the same content always call for the same format.

The goal of C-Series is not to select the correct output format. It is to make the process of output format selection observable as a structure.

Figure 1: C-Series Classification. The L2-C4 and L8-C4 examples are discussed in Chapter 8.



2-2. What Is Transmission Cost?

Transmission involves cost.

The sender carries the load of organizing content and producing it in some format. The receiver carries the load of decoding that output and arriving at meaning.

C-Series treats these collectively as transmission cost.

The critical point here is that optimizing transmission cost does not mean minimizing it.

C4 (Minimal Form) reduces the sender's output cost. But it can increase the receiver's decoding cost. C2 (Comprehension-first) reduces the receiver's decoding cost. But it increases the sender's output cost.

In other words, lowering one cost can raise another. Optimization in C-Series does not mean reducing cost to zero. It means selecting which cost to absorb, and to what degree, given the situation.

The optimal value of transmission cost shifts with context, recipient, and purpose. C-Series is a model that makes this selection observable as a structure.

2-3. Overview of C1 through C4

C-Series classifies output format into four types.

- C1: Structure-first
- C2: Comprehension-first
- C3: Reception-first
- C4: Minimal Form

C1 prioritizes the transmission of structure. As a result, sequence, causality, and reasoning tend to be made explicit. C2 prioritizes supporting comprehension. As a result, analogies, concrete examples, and restatements tend to be used. C3 prioritizes ease of reception. As a result, softening expressions and considerate phrasing tend to appear. C4 prioritizes reducing output cost. As a result, highly compressed or conclusion-only output tends to emerge.

These four types do not indicate superiority. Multiple formats may also be used simultaneously.

The key point is that even with identical content, the distribution of transmission cost shifts depending on output format. Observing which format is selected may make the structure of the transmission process more visible.

The details of each format are covered from Chapter 3 onward.

Chapter 3: C1 — Structure-First

C1 is an output format that prioritizes the transmission of structure.

In C1, thought content is organized into sequence, causality, and reasoning before output. The priority is enabling the receiver to grasp the structure of the content.

In this format, the logical arrangement of information serves as the primary basis for output design.

C1 tends to function well in situations where accurate transmission of content is required. Research presentations, technical explanations, and analytical reports are examples where this format is commonly observed.

C1 has its limits. The more explicitly structure is laid out, the greater the volume of information. This can increase the receiver's decoding cost. In emotionally unstructured contexts, C1 output may also become harder to receive.

The goal of C1 is not to provide the correct structure. It is to make the structure-first output tendency observable.

[Example] When outputting the content "this proposal will not be adopted" in C1, the result tends to take a form such as "We will not be adopting this proposal. The reasons are A, B, and C" — making the conclusion and reasoning explicit in sequence.

Chapter 4: C2 — Comprehension-First

C2 is an output format that prioritizes supporting comprehension.

In C2, the priority is enabling the receiver to understand the content more easily. To achieve this, analogies, concrete examples, and restatements are used to convert abstract content into a form the receiver can grasp.

In this format, reducing the receiver's decoding cost serves as the primary basis for output design.

C2 tends to function well in situations where a knowledge gap exists between sender and receiver. Explanations, education, and introductory-level guidance are examples where this format is commonly observed.

C2 has its limits. The more analogies and restatements are used to aid understanding, the more the accuracy of the original content can decrease. Excessive simplification may cause the core of the intended message to become unclear.

The goal of C2 is not to guarantee correct understanding. It is to make the comprehension-first output tendency observable.

[Example] When explaining gravity to a child, output tends to take a form such as "Things fall because the Earth is pulling them toward it" — converting the concept using familiar imagery and analogy. This is the output tendency of C2.

Chapter 5: C3 — Reception-First

C3 is an output format that prioritizes ease of reception.

In C3, the priority is enabling the receiver to accept the content more readily. To achieve this, softening expressions and considerate phrasing are used to reduce friction in the output.

In this format, prioritizing receptivity over content itself serves as the basis for output design.

C3 tends to function well in situations where the receiver's reaction has a greater influence on the outcome of transmission than the content itself. Negotiation, coordination, and relationship-oriented dialogue are examples where this format is commonly observed.

C3 has its limits. The more ease of reception is prioritized, the more information density can decrease. When softening expressions accumulate, the core of the intended message may become unclear.

The goal of C3 is not to guarantee smooth communication. It is to make the reception-first output tendency observable.

[Example] When outputting the content "this proposal will not be adopted" in C3, the result tends to take a form such as "This is an interesting proposal. We will keep it in mind for future reference" — converting the message into a form that reduces friction.

Chapter 6: C4 — Minimal Form

C4 is an output format that prioritizes reducing output cost.

In C4, the sender assumes a level of shared premise and transmits content in compressed form. To achieve this, highly compressed, conclusion-only, or highly abstract expressions tend to be used. Supplementation is left to the receiver.

In this format, output design is based on the assumption that the receiver will supply what is missing.

C4 tends to function well in situations where the sender can assume a high degree of shared premise. Dialogue between experts, time-critical judgment calls, and high-speed exchanges are examples where this format is commonly observed.

C4 has its limits. When the shared premise the sender assumed does not exist on the receiver's side, the receiver's decoding cost rises sharply. The risk of misunderstanding also increases.

The goal of C4 is not to guarantee efficient transmission. It is to make the output tendency of prioritizing output cost reduction observable.

[Example] When outputting the content "this proposal will not be adopted" in C4, the result tends to take a highly compressed form such as "Not viable" or "Difficult to proceed with."

Chapter 7: Examples

The preceding chapters have each addressed C1, C2, C3, and C4 individually. Chapter 7 examines what changes when the same content is delivered in four different output formats.

Situation

In the context of relationship advice, the content to be communicated is: "Things will end if they stay this way."

C1 Output

"The way things are going, I think there's a high likelihood it will end — because of A, B, and C."

Structure is made explicit. Conclusion and reasoning are delivered in sequence. The receiver can follow the logic of the content.

C2 Output

"The way things are going, they might end up resenting you."

The abstract notion of relationship deterioration is converted into a form the receiver can more readily picture. The word "break up" does not appear.

C3 Output

"Maybe try working on getting them to like you more?"

Friction is reduced through softened phrasing. The core of the message does not surface in the output.

C4 Output

"It's over."

The sender assumes shared premise and delivers a compressed output. Supplementation is left to the receiver.

Across the four formats

Faced with the same content, C1 prioritizes structure, C2 prioritizes situational understanding, C3 prioritizes reception, and C4 prioritizes compression. It can be observed that the distribution of transmission cost differs across formats.

C-Series is a model that makes these differences observable.

Chapter 8: LLM and C-Series

We communicate something to someone every day. So do LLMs.

Current LLMs likely select some form of output format when generating responses. However, which format tends to be selected may vary depending on the model and context.

In observational terms, C1 or C2 output tends to be selected in situations requiring structural explanation or information delivery. On the other hand, in situations where the user is in an emotional state, or in models designed with an emphasis on empathy, C3-like output is sometimes observed.

In any case, C-Series is a framework for observing output format only. How LLMs internally select output format falls outside the observational scope of C-Series.

The key point is that even within the same C format, a difference in origin changes the meaning. For example, a child's "WAHHH!" and a considered "No way" may both resemble C4 as output formats. Yet their origins differ. C-Series can observe differences in C format, but identifying the origin layer requires SCOPE.

C-Series may function as a framework for observing the output format of LLMs.

Chapter 9: Connection to the COS Series

C-Series is a model that organizes the process of output format selection into an observable structure. However, C-Series does not stand alone.

In actual cognitive processing, the selection of output format functions within an interaction with other processing systems. Interpretation, decision-making, and observation exist in parallel with C-Series, or before and after it. COS is a framework that integrates these processing systems with observation through SCOPE, and C-Series constitutes one part of that framework.

B-Series sits upstream of C-Series. Where B-Series addresses how we receive what others say, C-Series addresses how we output what we intend to say. The result of interpretation may influence the selection of output format.

A-Series may function in parallel with C-Series. Where A-Series addresses what to select, C-Series addresses how to convey it. How the result of a decision is delivered in output form becomes an observation target for C-Series.

C-Series can observe C formats, but cannot identify the origin layer. Even within the same C4, the meaning differs depending on which origin layer generated it. Identifying the origin layer requires SCOPE.

Within COS, the A, B, and C series function in parallel and are integrated with observation through SCOPE, making the full picture of cognitive processing readable. Within that structure, C-Series is positioned as the processing system responsible for observing output format.

COS also includes D-Series, which serves as the observation and verification system supporting the A, B, and C processing systems. D-Series will be addressed in the next paper.